Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/31/2006
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Project Title:	Novel Polymers for Bioreactor Based Bone Tissue Engineering		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22903	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Levine, Elliot (The Wistar Institute) Nair, Lakshmi (University of Virginia, Department of Orthopedic Surgery)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	This research is aimed at evaluating the synergistic affects between novel polyphosphazene scaffolds and NASA High Aspect Ratio Vehicle (HARV) bioreactors in the field of bone tissue engineering. The research aims follow the central hypothesis that novel biocompatible buoyant scaffolds can be developed from biocompatible and biodegradable polyphosphazenes and 3D dynamic flow culture conditions in the rotating NASA bioreactor would enhance bone cell phenotype development and matrix synthesis in biodegradable polyphosphazene scaffolds in vitro. The polyphosphazene scaffolds are very advantageous for tissue engineered constructs due to their hydrolytic instability, non-toxic and neutral degradation products, ease of matrix fabrication, biocompatibility and osteoconductivity. These unique scaffold properties do not create the acidic degradation products of PLAGA scaffolds that can lead to inflammatory responses in vivo. We anticipate that these novel polyphosphazene based tissue engineered 3D scaffolds developed under dynamic flow environment will be able to address many challenges in bone tissue engineering.		

Task Description:	Our objectives are to identify and synthesize novel biodegradable and biocompatible polyphosphazenes as potential candidates for bone tissue engineerign application, fabricate porous three dimensional buoyant strucutres from biocompatible polyphosphazenes and to determine the effects of 3D dynamic flow conditions created by the rotating NASA bioreactor on osteoblast cells cultured in biodegradable 3D polyphosphazene scaffolds in terms of cell proliferation, phenotype expression and mineralization in vitro. It has been found that bone loss associated with space flight can result in additional risk of fracture and impaired bone healing. Threat to the crew health and therefore the mission depend on fracture site, severity and treatment options available. Countermeasure research and development of novel technologies to address musculoskeletal injury is therefore very important. Our research to address this issue is focused on engineering a viable tissue substitute for musculoskeletal regeneration. Musculoskeletal tissue engineering is currently developing as one of the most promising therapeutic strategy to treat musculoskeletal injuries. We adopt a multi-plex approach by combining biodegradable three dimensional scaffolds having appropriate mechanical properties, cells and dynamic culture conditions using rotating NASA bioreactor to engineer viable tissue substitutes. To fabricate scaffolds with optimum physical and biological properties we are developing a novel class of biodegradable polymer, polyphosphazenes, with neutral and non-toxic degradation products and high biocompatibility. This study will enable us to identify novel fabrication techniques suitable for developing buoyant porous strucutres from biodegradable and biocompatible polyphosphazenes and the critical variables that affect bone cell response and gene expression which influence the quantity, composition and organization of the resulting synthesized matrix which eventually impact the quality of the bone and therefore the success of tissue engineered bone implants in vivo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research led to the development of some candidate biocompatible and biodegradable polyphosphazenes a novel class of biomaterial for bone tissue engineering applications. The research also led to the development of novel fabrication technique that can be used to form porous buoyant strucutres from biodegradable polyphosphazenes as well as other polymers as scaffolds for tissue engineering. A provisional patent has been filed on the novel technology. The effects of 3D dynamic flow conditions created by the rotating NASA bioreactor on osteoblast cells cultured in biodegradable 3D buoyant polyphosphazene scaffolds in terms of cell proliferation, phenotype expression and mineralization in vitro is undergoing. The results of these studies will enable us to identify a novel class of polymer that can support accelerated tissue regeneration to address musculoskeletal tissue injury.
	During the past two years, we were focusing on specific aim #1 and part of specific aim #2 in evaluating the biocompatibility of novel biodegradable polyphosphazenes developed in the first year and developing methods to fabricate porous, three dimensional ligher than water matrices from biodegradable polyphosphazenes with appropriate physical and mechanical properties for bioreactor based tissue engineering. The formed matrices were compared with matrices prepared using biodegradable poly(lactide-co-glycolide) with lactide-glycolide ratio of 85:15. (PLAGA 85:15). Biocompatibility evaluation of biodegradable polyphosphazenes: We have developed homo and mixed substituent biodegradable amino acid ester polyphosphazenes based on alanine ethyl ester. Two additional homopolymers have been synthesized as well--poly[bis(ethyl phenylalaninato)phosphazene]; PNEPhA, and poly[bis(methyl valinato)phosphazene], PNMV. The polymers were synthesized by the two step macromolecular substitution process as described in the earlier report. The synthesized polymers were characterized by multinuclear magnetic resonance spectroscopy (NMR), Differential scanning calorimetry (DSC), and Gel permeation chromatography (GPC). The biocompatibility of the PNEA and PNEAPhPh was determined using in vitro and in vivo techniques in our laboratory recently. The in vitro biocompatibility was determined by following the adhesion and proliferation of primary rat osteoblast cells on thin films of the synthesized polymers and compared with a slow degrading polymer PLAGA 85:15. Polymer Film Preparation: The polymer solution was prepared by separately dissolving 0.7 g of polyphosphazenes (PNEAmPh with EA:mPH ratio of 50:50 and 75:25) and PLAGA in dichloromethane. Cell Viability and Proliferation Primary rat osteoblast (PRO) cells were isolated from calvaria of neonatal Sprague-Dawley rats according to standard procedure. The matrices were sterilized by UV irradiation and seeded with cells and the incease in cell number was followed by a standard biochemical assay (MTS assay). Determination of bioactivity of polyphosphazenes: The bioactivity, i.e., the efficacy of polyphosphazene to nucleate hydroxyapatite (HAp) on the surface as an indication of osteointegritivity of the materials was evaluated. by incubating the matrices in simulated body fluid and following HAp formation on the suface via SEM. Fabrication of biodegradable polyphosphazene microspheres (PNEAmPh, PNEAPhPh, PNEA, and PNEPhA): The PNEAmPH and PNEAPhPh microspheres were prepared by a standard solvetn evaporation method. Briefly, PNEAmPh was dissolved in methylene chloride at 7% w/v. The polymer solution was added as a thin steady stream approximately 6 in from the surface into 1000 ml solution of 1% poly(vinyl alcohol), PVA, in deionized water. The solution was stirred at a rate of 400 rpm during addition of the polymer solution. Once the addition was complete the stirring speed of the solution was increased to 600 rpm and stirred overnight to allow the evaporation of the methylene chloride. The PNEA microspheres were prepared as follows: PNEA was dissolved in methylene chloride at 9% w/v, and poured into a solution of 1% PVA stirred at 400 rpm and initially chilled to 5oC. The PNEPhA microspheres were prepared as follows: PNEPhA was dissolved in methylene chloride at 24% w/v, and poured into a stirred solution of 1% PVA stirred at 400 rpm and initially chilled to ~5oC. The microspheres were then isolated from the solution using vacuum filtration. After isolation the microspheres were lyophilized for an additional 48 hours to ensure that there was no residual moisture or solvent. Fabrication of porous, lighter than water, microspheres (PNEA, and PNEPhA): PNEA and PNEPhA porous microspheres were prepared as follows: PNEA was dissolved in methylene chloride at 9% w/v, 10% v/v hexane, and 10% v/v water was added to the polymer solution. The resulting mixture was vortexed for 30 seconds to create a suspension. The polymer suspension was then poured into a 1% solution of PVA stirred at 400 rpm. Likewise PNEPhA microspheres were prepared as follows: PNEPhA was dissolved in methylene chloride at 24% w/v, 10% v/v hexane, and 10% v/v water was added to the polymer solution. The resulting mixture was vortexed and poured as stated above.

Fabrication of three dimensional matrices from biodegradable polyphopshazene microspheres: A novel solvent/non-solvent system was used to sinter the biodegradable polyphosphazene microspheres together to form the porous three dimensional matrices. A mixture of acetone and hexane was used as the solvent/non-solvent system. The ratios of acetone hexane used was dependant on the average molecular weight of the polymer used as well as the desired amount of bonding needed to form the three-dimensional matrices. Briefly it has been found that high molecular weight polyphosphazene will require more solvent to create the same amount of bonding. Like wise more solvent will increase bonding in similar molecular weight polymers. The three dimensional matrices were prepared as follows. The dried microspheres were poured into a cylindrical mold. Approximately 1 microliter of solvent/non-solvent solution was added for a microliter of the mold volume. The solvent was then allowed to evaporate for 30 min in a fume hood. The matrices in the mold were then dried under vacuum for an hour. The mold was then cooled in liquid nitrogen for 10 min followed by thawing. The process was repeated 3 times to loosen the scaffolds from the sides of the mold. The scaffolds were then taken out of the mold and lyophilized for 48 hours.

The above technique was modified to allow a more consistent sintering of the spheres throughout the volume of the scaffold. This is accomplished by creating a slurry with the microspheres and the sintering solution, and then molding the slurry and allowing it to dry as mentioned above. Briefly, PNEA microspheres (both solid and hollow) are combined with a 17.5% - 22.5% v/v mixture of acetone and hexane, approximately 4 microliters of sintering solution per milligram of polymer is used. The resulting slurry is vortexed for 5 sec and molded into the same mold as before. After solvent evaporation the scaffold is demolded as before. PNEPhA microspheres are sintered in a similar way with a 30% - 40% v/v mixture of THF and hexane.

Nanofiber Incorporation: PLLA was dissolved in N,N dimethylformamide heated to 100oC and stirred to create a 1.5% (wt/vol) solution. The solution is then pipetted over the preexisting microsphere structures in a mold of arbitrary shape with a closed bottom. The mold is transferred to a -80oC freezer for 24 hours to allow gelation of the PLLA/DMF solution. The new constructs are first transferred to hexane maintained at -20oC for 24 hours and then to room temperature DI water for 24 hours to allow removal of DMF and subsequent precipitation of PLLA fibers. After the liquid/liquid solvent removal the constructs are lyophilized to ensure total solvent removal.

Task Progress:

RESULTS AND DISCUSSION

The structure of the synthesized polymers was confirmed by multinuclear NMRs.

Osteoblast cell viability and proliferation on PNEA50mPh50 and PNEA75mPh25 discs were assessed using the MTS assay and compared to PLAGA85:15 films and TCPS. The cell numbers on the polymeric discs did increase with time demonstrating that the cells were able to proliferate on the polyphosphazenes.

These studies demonstrated the high osteocompatibility of polyphosphazenes making them ideal candidates for bone tissue engineering applications.

The bioactivity of polyphosphazenes was determined by incubating polyphosphazene having carboxyl side groups in simulated body fluid. The SEM evaluation showed the formation of mineral on the surface of the polymer film, the thickness of which increase with time of incubation. The study demonstrated the feasibility of developing polyphoshazenes having intrinsic bioactivity (osteointegritivity) for musculoskeletal applications.

Polyphosphazene microspheres were prepared by solvent evaporation method. The study demonstrated the feasibility of developing biodegradable polyphosphazene microspheres by the above described method.

Different techniques were evaluated to develop porous three-dimensional matrices from biodegradable polyphosphazene microspheres. Heat sintering the microspheres, the technique that has been developed in our laboratory for fabricating PLAGA scaffolds was not effective for developing polyphosphazene matrices presumably due to the lower Tg of biodegradable polyphosphazenes compared to PLAGA. Another method investigated was using a dilute solvent sintering method using a dilute solution of acetone to fuse the microspehres together to form the porous matrices. However, even at very high dilution, the solvent system led to almost complete fusion of the microspheres rather than connecting the spheres together to form a porous matrix. So a novel approach has been developed using solvent/non-solvent sintering system, where in a mixture of acetone (solvent) and hexane (non-solvent) was used to sinter the microspheres. The presence of non-solvent prevented the complete dissolution/fusion of the microspheres and resulted in porous matrices.

The study demonstrated that increasing the concentration of acetone has a significant effect on the extent of bonding between the spheres. As you increase the concentration of acetone the extent of bonding between the microsphres increased as demonstrated by the SEMs.

The SEM evaluation was performed with PNEAPhPh microsphere matrices formed by solvent/non-solvent method. Thus SEMs showing the sintered porous matrices of PNEA50PhPh50 using a solvent non-solvent system composed of 5% acetone and 95% hexane, 10% acetone and 90% hexane. and 15% acetone and 85% hexane were investigated.

In the case of PNEAPhPh scaffolds, the trend was more significant and as shown in the SEMs, increase in acetone concentration significantly increased the connectivity of the microspheres.

To quantify the resulting microsphere scaffolds and determine to what degree the porosity is maintained throughout the volume of the scaffold mercury porosimetry was used. The results show a total porosity of 31.92 ± 0.385% and a median pore size of 179.13 ± 2.047 µm, both of which are very similar to values reported earlier for optimized heat sintered PLAGA microsphere scaffolds. This suggests the solvent/non-solvent sintering method produces an end result very similar to that of the more conventional heat sintering for PLAGA microspheres. Furthermore it suggests that the porosity and interconnectivity of that porosity remains constant throughout the volume of the scaffold, which is a critical design consideration when creating a scaffold for tissue engineering.

Our next attempt was to develop lighter than water porous microsphere scaffolds from biodegradable polyphosphazenes as potential candidates for bioreactor based bone tissue engineering applications. Our previous studies have shown that lighter than water poly(lactide-co-glycolide) microsphere scaffolds provide a means of culturing cells in a bioreactor environment at near physiological levels of shear. In addition, cell culture in a bioreactor provides better nutrient transfer/waste removal from the scaffold allowing larger structures to be cultured than would be possible in a static environment limited to diffusion in just one direction.

Porous microspheres (ligher than water) were fabricated from both PNEA and PNEPhA as described earlier. Different

	ratios of LTW and HTW micropsheres were mixed to develop scaffolds having the target density of target 0.85 g/ml. They were verified to be lighter than water by first immersing the dry microspheres in a stirred solution of DI water for 24 hours. After 24 hours the microspheres remained buoyant. The intact porous structure of the PNEPhA microspheres were verified by SEM. The cross section of the porous sphere appears to contain many non-connected pores, which would explain the buoyancy retention after immersion in the DI water. Since nanofibrous structures are known to significantly improve cell adhesion and proliferation based on our previous studies using poly(lactide-co-glycolide), attempts were made to develop the most optimized scaffolds for tissue engineering by combining solvent-nonsolvent sintering with phase separation. Our studies demonstrated the feasibility of developing porous sintered polyphosphazene microsphere matrices impregnated with polylactide nanofibers. A series of solvents were tested based on their ability to form a thermally induced gel with poly(L-lactide), as well as whether they interrupted the already formed polyphosphazene microsphere matrix. Two solvents were found to be moderately effective; dimethylformamide, and acetone. However, acetone was not nearly as effective at producing nanofibers from the poly(L-lactide) and also would begin to dissolve the microsphere matrix if given sufficient time. The studies described here enabled for the first time to identify a suitable fabrication method to develop porous three-dimensional matrices from biocompatible polyphosphazenes. This optimized density of the microsphere scaffolds (of 0.85 g/ml) allows for the scaffold to rotate in a bioreactor with a steady orbit, whereas scaffolds too heavy/light result in numerous collisions between the scaffold and the bioreactor wall potentially disrupting cell growth. These porous scaffolds with micro-nanostructures is currently been evaluated to determine the effect of dynamic and static culture conditions on cellular proliferation and differentiation.
Bibliography Type:	Description: (Last Updated: 10/08/2019)
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Patents	Pending. Provisional patent filed August 2006. Aug-2006 Brown JL, Nair LS, Laurencin CT. "Solvent/Non-Solvent Sintering Technique for Creating Sintered Microsphere Scaffolds."